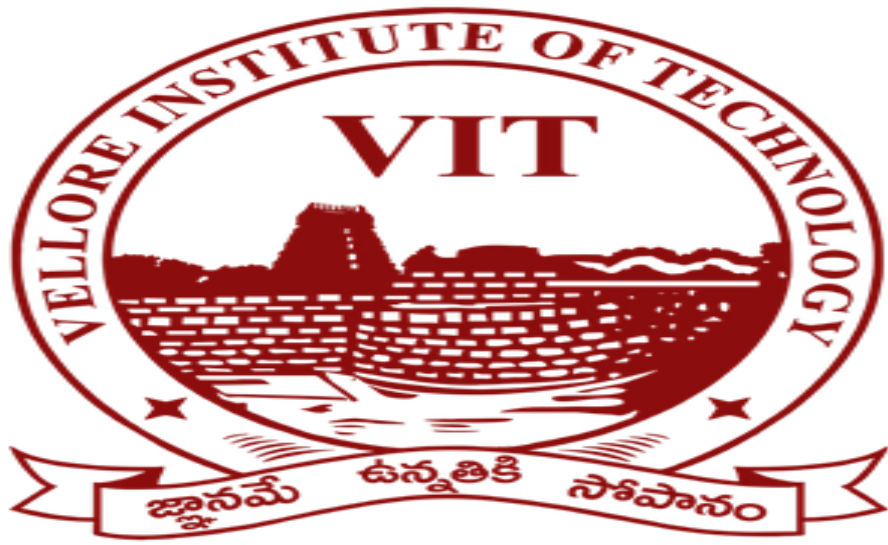




RFID BASED SMART ATTENDANCE

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1. INTRODUCTION

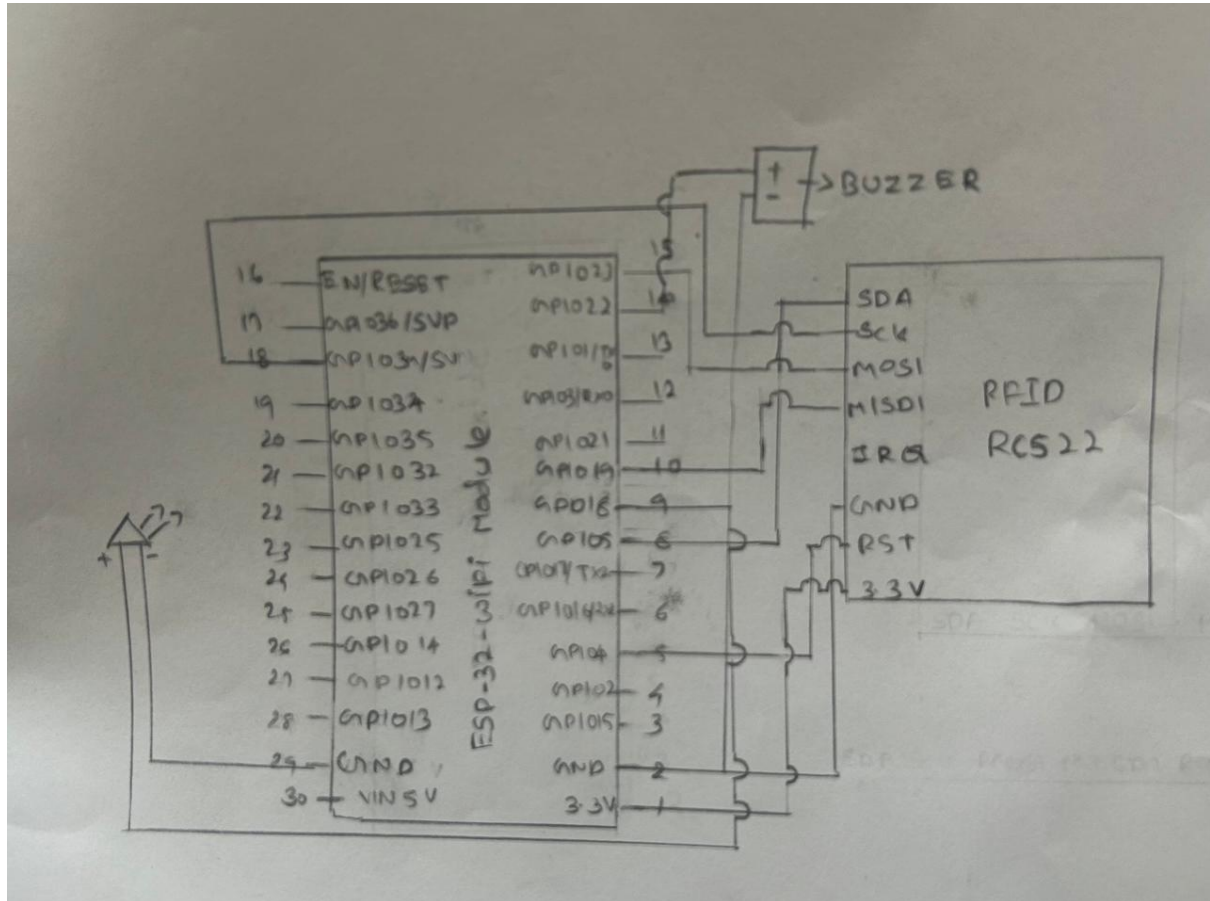
Traditional attendance methods like paper registers and roll calls waste time and allow proxy marking, creating frustration for students and teachers alike. These manual processes disrupt learning and are prone to human error, highlighting the need for a faster, more reliable system.

An RFID-based smart attendance system solves this by letting individuals mark their presence with a simple tap of a card, taking less than a second. This touch-free, automated approach saves time, prevents fraud, and allows educators to focus on teaching rather than tracking.

2. COMPONENTS

- ESP 32
- RFID RC522 MODULE
- JUMPER WIRES
- BREADBOARD
- RFID CARD

3. CIRCUIT DIAGRAM



4. CODE

ESP 32 Code :

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <SPI.h>
#include <MFRC522.h>

#define SS_PIN 5
#define RST_PIN 4
```

```
#define LED_PIN 26

#define BUZZER 25


bool useLED = false;
bool useBuzzer = false;


MFRC522 mfrc522(SS_PIN, RST_PIN);

const char* ssid = "cheta laptop";
const char* password = "cheta@005";

String serverName =
"https://script.google.com/macros/s/AKfycbywR9ag0mnVQwTGu0hYdIKRkualu7-
aNTP0-oo_aX3RnrlATxGtFeC3OOS8s6GXISkAgA/exec";

String validUID = "DF 72 A8 16";


void setup() {
  Serial.begin(115200);


  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER, OUTPUT);


  SPI.begin();
  mfrc522.PCD_Init();


  WiFi.begin(ssid, password);
  Serial.print("Connecting WiFi");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
}
```

```
Serial.println("\nWiFi Connected");  
Serial.println("Scan RFID card...");  
}
```

```
void beep() {  
    if (!useBuzzer) return;  
  
    digitalWrite(BUZZER, HIGH);  
    delay(200);  
    digitalWrite(BUZZER, LOW);  
}
```

```
void sendToGoogle(String name, String status) {  
    if (WiFi.status() == WL_CONNECTED) {  
        HTTPClient http;  
  
        String url = serverName + "?name=" + name + "&status=" + status;  
        http.begin(url);  
        http.GET();  
        http.end();  
    }  
}
```

```
void loop() {  
  
    if (!mfrc522.PICC_IsNewCardPresent()) return;  
    if (!mfrc522.PICC_ReadCardSerial()) return;
```

```
String uid = "";
for (byte i = 0; i < mfrc522.uid.size; i++) {
  if (mfrc522.uid.uidByte[i] < 0x10) uid += "0";
  uid += String(mfrc522.uid.uidByte[i], HEX);
  if (i != mfrc522.uid.size - 1) uid += " ";
}

uid.toUpperCase();

Serial.print("UID: ");
Serial.println(uid);

beep();
if (uid == validUID) {
  Serial.println("Access Granted");

  if (useLED) digitalWrite(LED_PIN, HIGH);

  sendToGoogle("23BEC7275", "Present");

  delay(3000);

  if (useLED) digitalWrite(LED_PIN, LOW);
}

delay(2000);
mfrc522.PICC_HaltA();
}
```

AppScript Backend Server:

```
var ss = SpreadsheetApp.openById('1NZ8tbYmEz1ncT4Zs_URjSaN-060--zyIYVViOLHxvmM');
```

```
var sheet = ss.getSheetByName('Sheet1');
```

```
var timezone = "asia/dhaka";
```

```
function doGet(e){
```

```
  Logger.log( JSON.stringify(e) );
```

```
  if (e.parameter == 'undefined') {
```

```
    return ContentService.createTextOutput("Received data is undefined");
```

```
  }
```

```
  var Curr_Date = new Date();
```

```
  var Curr_Time = Utilities.formatDate(Curr_Date, timezone, 'HH:mm:ss');
```

```
  var name = stripQuotes(e.parameters.name);
```

```
  var nextRow = sheet.getLastRow() + 1;
```

```
  sheet.getRange("A" + nextRow).setValue(name);
```

```
  sheet.getRange("B" + nextRow).setValue(Curr_Date);
```

```
  sheet.getRange("C" + nextRow).setValue(Curr_Time);
```

```
  return ContentService.createTextOutput("Card holder name is stored in  
column A");
```

```
}
```

```
function stripQuotes( value ) {
```

```
  return value.toString().replace(/^["]|["]$/g, "");
```

```
}
```

```
function doPost(e) {
```

```

var val = e.parameter.value;

if (e.parameter.value !== undefined){
    var range = sheet.getRange('A2');
    range.setValue(val);
}
}

```

5. PINOUT

RC 522	ESP 32
3.3 V	3.3 V
GND	GND
SDA	D 5
SCK	D 18
MOSI	D 23
MISO	D 19
RST	D 4

6. EXPLANATION

In many institutions, attendance is still taken manually, which takes time and can lead to mistakes. To solve this issue, we developed an RFID-based attendance system that automates the process. Each user is provided with an RFID card that contains a unique identification number. When the card is scanned, the system quickly reads and processes the data.

The attendance is then automatically updated to an Excel sheet. This reduces manual work and avoids errors caused by manual entry. The system also allows real-time tracking, making it easy to monitor attendance. Overall, it improves efficiency, accuracy, and saves time.

7. APPLICATIONS

In educational institutions, the system automates student attendance tracking during lectures, exams, and library visits, reducing teacher workload. In corporate offices, it records employee check-ins and check-outs seamlessly, integrating with payroll systems for accurate salary processing.

Hospitals use RFID attendance to monitor staff shifts and track trainee doctors during emergency rotations. Event management companies apply this technology to register participants at workshops or conferences, ensuring secure and swift entry.

8. CONCLUSION

The RFID-based smart attendance system automates tracking, eliminates manual errors and proxy marking, and saves time. It works effectively in schools, offices, hospitals, and events for fast, touch-free attendance. This project reduces administrative workload, allowing people to focus on more meaningful tasks.

Challenges Faced:

The main challenges were hardware interfacing, WiFi reliability, and proper data handling between the RFID system and cloud platform.